

12.08.26

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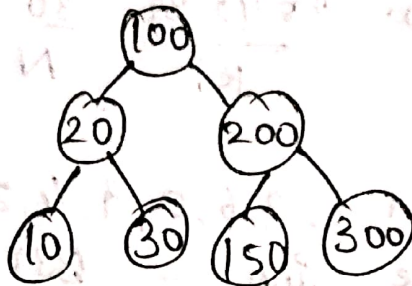
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CSA0311 DATA STRUCTURES

CO3 - AT3 - TREE TRAVERSAL

CHALLENGES

1.



Inorder:

LNR $\Rightarrow$ $\frac{10 \ 20 \ 30}{L}$ $\frac{100}{N}$ $\frac{150 \ 200 \ 300}{R}$

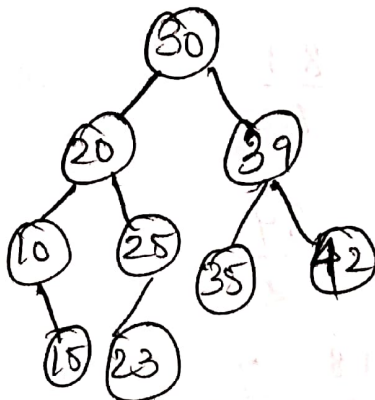
Preorder:

NLR $\rightarrow$ $\frac{100}{N}$ $\frac{20 \ 10 \ 30}{L}$ $\frac{200 \ 150 \ 300}{R}$

Postorder

LRN $\Rightarrow$ $\frac{10 \ 30 \ 20}{L}$ $\frac{150 \ 300 \ 200}{R}$ $\frac{100}{N}$

2.



Inorder

LNR $\Rightarrow$ $\frac{10 \ 15 \ 20 \ 23 \ 25}{L}$ $\frac{30}{N}$ $\frac{35 \ 39 \ 42}{R}$

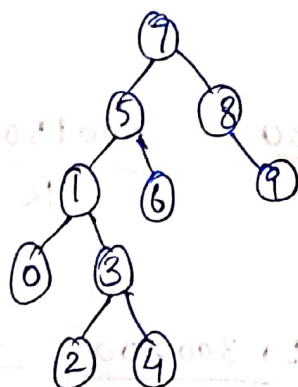
Preorder

NLR $\Rightarrow$ $\frac{8020}{N} \frac{10152523}{L} \frac{393542}{R}$

Postorder

LRN $\Rightarrow$ $\frac{1810232520}{L} \frac{354239}{R} \frac{30}{N}$

3. Assume the digits 7, 5, 1, 8, 3, 6, 0, 9, 4, 2 are added in that sequence into a BST that is initially empty. The binary search tree follows the standard natural number ordering. What is the resulting tree's traversal sequence?

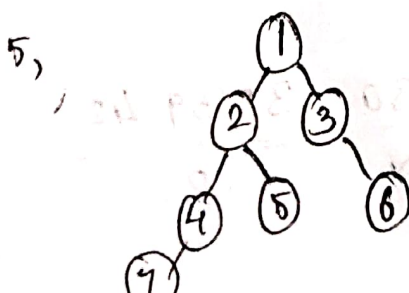


Inorder:

INR $\Rightarrow$ $\frac{0123456}{L} \frac{7}{N} \frac{89}{R}$

Preorder $\Rightarrow$ $\frac{7}{N} \frac{5103246}{L} \frac{89}{R}$

Postorder $\Rightarrow$ $\frac{0243165}{L} \frac{98}{R} \frac{7}{N}$

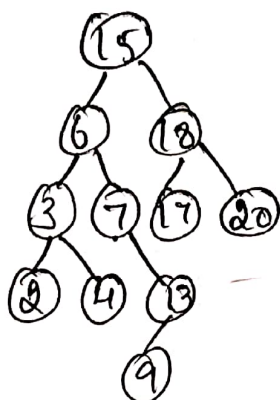


Preorder $\Rightarrow$ $\frac{1}{N} \quad \frac{2475}{L} \quad \frac{36}{R}$

Postorder $\Rightarrow$ $\frac{7452}{L} \quad \frac{63}{R} \quad \frac{1}{N}$

Inorder $\Rightarrow$ $\frac{7425}{L} \quad \frac{136}{N \quad R}$

6,

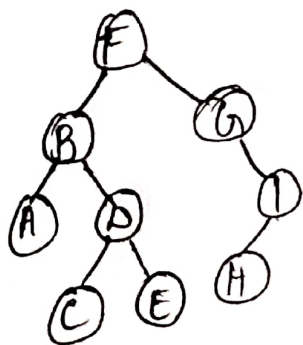


Inorder $\Rightarrow$ $\frac{23467913}{L} \quad \frac{15}{N} \quad \frac{171820}{R}$

Postorder $\Rightarrow$ $\frac{15}{N} \quad \frac{63247139}{L} \quad \frac{181720}{R}$

Preorder $\Rightarrow$ $\frac{24391376}{L} \quad \frac{172018}{R} \quad \frac{15}{N}$

7,



(c) Inorder $\Rightarrow$ $\frac{A B C D E}{L} \quad \frac{F}{N} \quad \frac{G H I}{R}$

8. Inorder - DBEFA~~G~~HLC
 Preorder - ABDEFC~~G~~HA

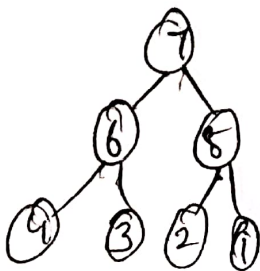
What is the postorder traversal of the same tree

~~Order~~:

Postorder:

(a) DFE~~B~~HLC~~A~~

9.



Inorder:

$$LNR \Rightarrow \frac{4637}{L \quad N} \quad \frac{251}{R}$$

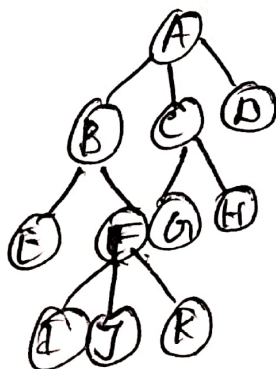
Preorder -

$$NLR \Rightarrow \frac{7}{N} \cdot \frac{643}{L} \cdot \frac{521}{R}$$

Postorder:

$$LRN \Rightarrow \frac{436}{L} \quad \frac{215}{R} \quad \frac{7}{N}$$

10.



Preorder: ABE~~F~~IJKCHAD

Postorder: E~~I~~JKFBGHCDA

Inorder: Not Applicable for a general tree